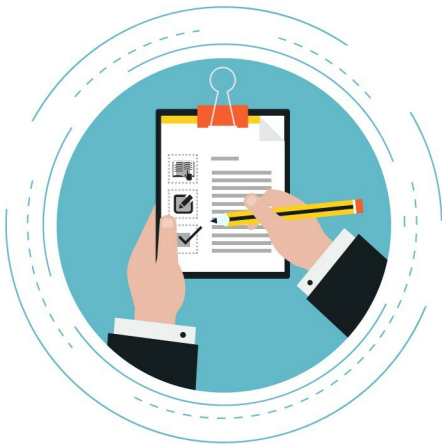


Enjoy Better Flexibility in Granting File System Permissions *with* Access Control Lists (ACLs)

ACLs provide additional and more flexible permissions for file systems. They give better control over who can read, write and execute a file or any disk resource.



Standard Linux file permissions are satisfactory for most situations but they have their limitations. Permissions can be set to restrict access to a file to only the file owner, to members of a single group, or to everyone else. It may not be appropriate for the process (a running program) to be a member of the group owning the files, and even less desirable to grant permission to everyone.

ACLs allow fine-grained permissions to be allocated to a file. Named users or named groups, as well as users and groups identified by a UID or GUID, can be granted permissions, in addition to the standard file owner, group owner and other file permissions. The same permission flags apply: *r* – read, *w* – write, and *x* – execute (on files, search for directories).

The file owner can set ACLs on individual files or directories. New files and sub-directories can automatically inherit ACL settings from the parent directory's default ACLs, if they are set. Just like normal file access rules, the parent directory hierarchy will need at least the *other execute* permission set to enable named users and named groups to have access.

The file system mount option

The file system needs to be mounted with ACL support enabled. XFS file systems have built-in ACL support. Ext4 file systems created on Red Hat Enterprise Linux 7 (RHEL 7) have the ACL option enabled by default, but ext4 file systems created in earlier versions of RHEL may need the ACL option